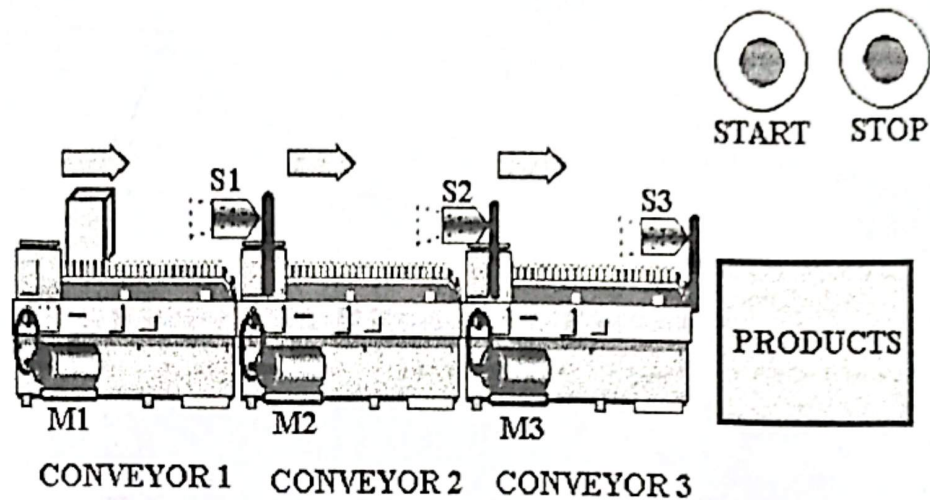


EXPERIMENT # 16

LADDER LOGIC PROGRAMMING: CONVEYOR BELT SERIES MOTOR CONTROL

Conveyor Belt Series Motors Control



▪ Determination of Operation Logic:

This is a simple motion control of conveyor belts. When the "START" button is pressed the Motor "M1" will start and the product will come out on the conveyor 1. When the product reaches the sensor "S1" and the product is detected then Motor "M2" will start and the Motor "M1" will stop. When the product reaches sensor "S2" then the Motor "M3" will start and Motor "M2" will stop. When sensor "S3" detects the product then Motor "M1" will again start and the Motor "M3" will stop and the above sequence will be repeated until stopped by the "STOP" button. The PT Screen can show the "M1", "M2" and "M3" ON & OFF indication and the operation can be "Started" and "Stopped" from the Screen Keys.

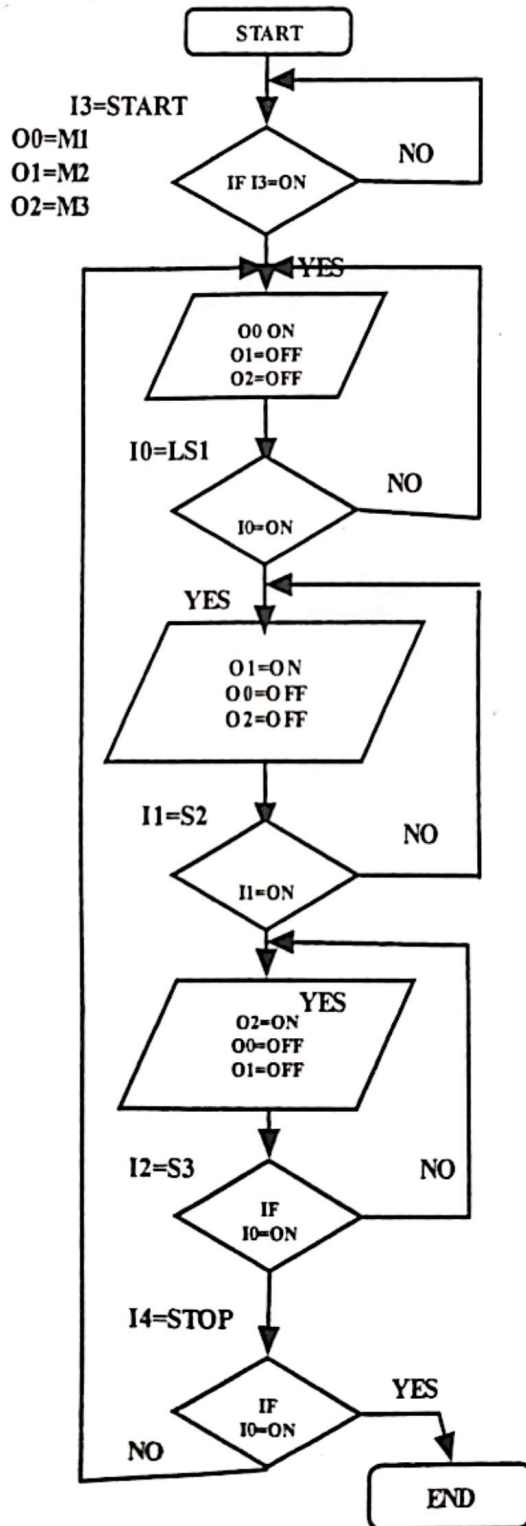
▪ Input/Output Address Declaration:

INPUTS		OUTPUTS	
Address	Description	Address	Description
I0	Sensor S1	O0	Motor M1
I1	Sensor S2	O1	Motor M2
I2	Sensor S3	O3	Motor M3

BASIC PLC KIT

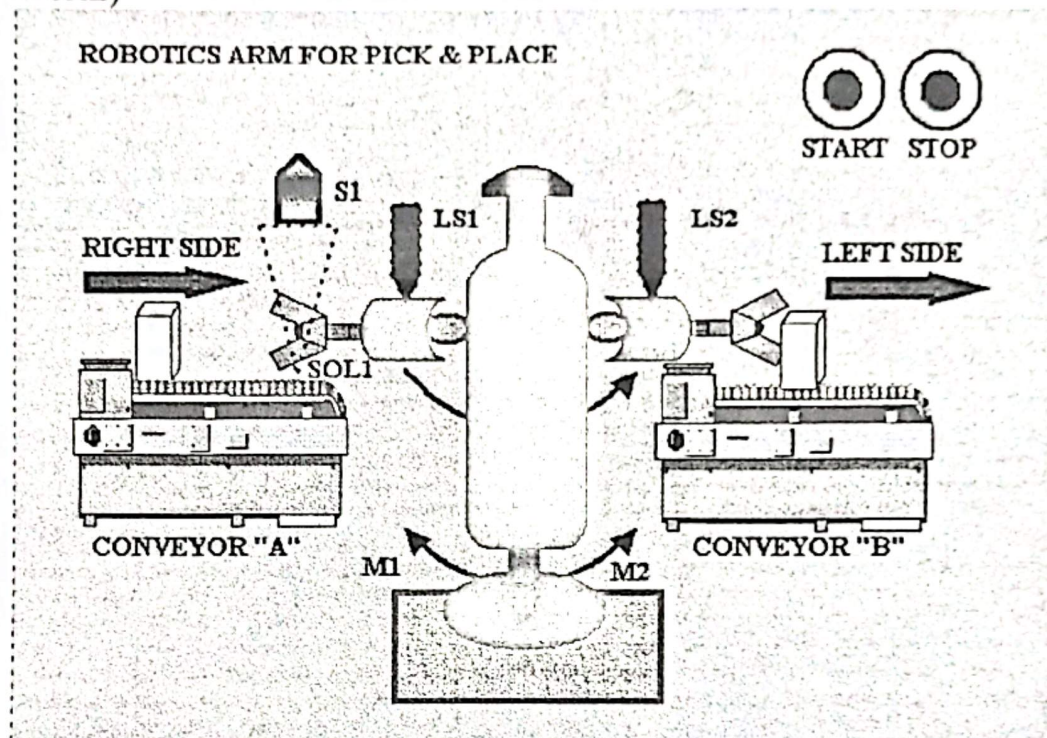
I3	Start		
I4	Stop		

Logic Flowchart:



EXAMPLE#10

Robotic Arm's Control (For Pick and Place Works)



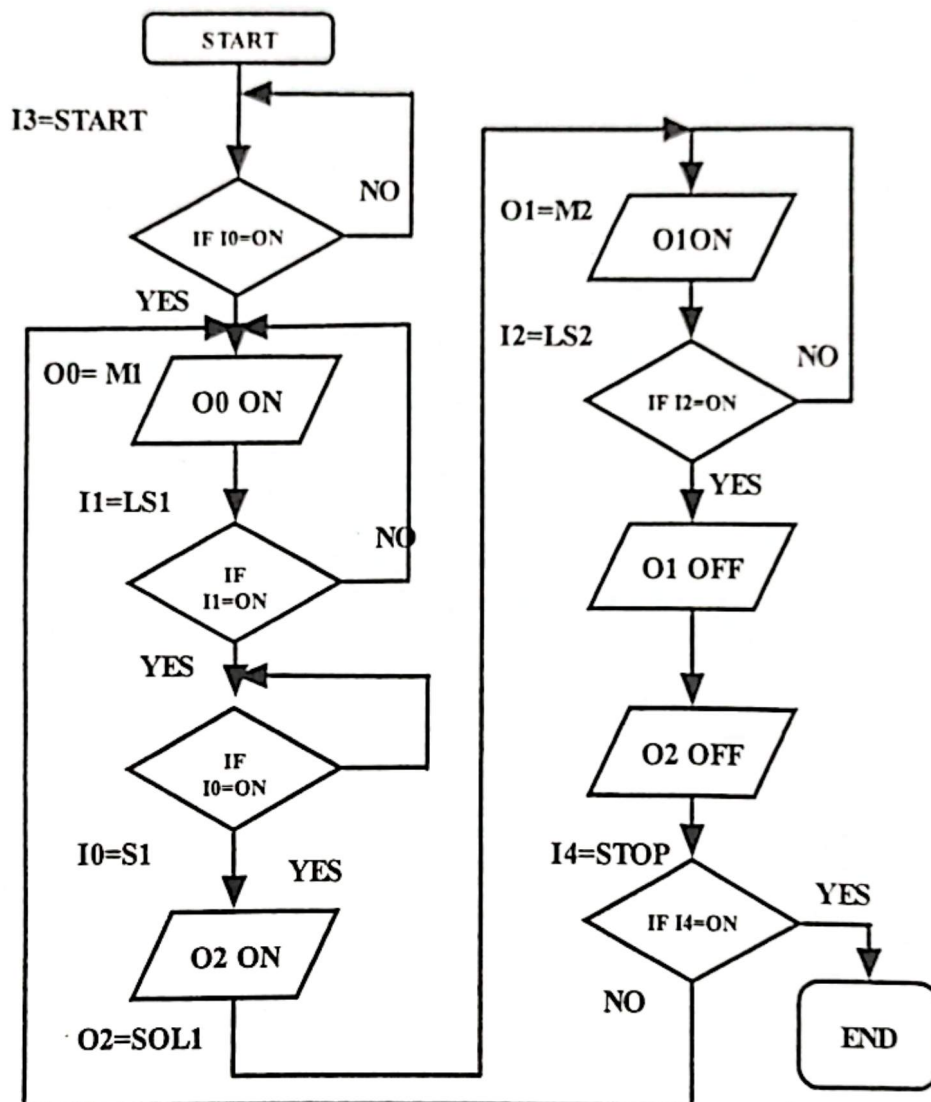
▪ Determination of Operation Logic:

A simple robot to show pick and place function. As the figure shows that there are two conveyor belts "A" and "B", products from the conveyor "A" are picked and placed on the conveyor "B". Pressing the "START" button will start the operation of the above Machine. The robot rotates its Arm to Right Side through the Motor M1 and when the Arm reaches to the position of work on the Conveyor "A", the limit switch LS1 will stop the M1. When the sensor S1 senses the product the SOL1 will be activated and the Arm will grasp the product and rotate anticlockwise to the Left Side through the Motor M2. When the Arm reaches the limit switch LS2, the M2 will stop. The product will be placed on the Conveyor "B" by releasing SOL1. The Arm will be again rotated clockwise to the Right Side and the above process will be repeated until stopped by pressing the "STOP" button.

The above operation can be controlled by the PT Screen. The "START" and "STOP" buttons can be controlled by the Screen Keys. The indication of Solenoid

valves can be shown on the Screen. The whole process of the Robotics Arm can be shown on the PT Screen step by step.

Logic Flowchart:



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- **Input/Output Address Declaration:**

INPUTS		OUTPUTS	
Address	Description	Address	Description
I0	Sensor S1	O0	Motor M1
I1	Limit Switch LS1	O1	Motor M2
I2	Limit Switch LS2	O2	Solenoid SOL1
I3	Start Button		
I4	Stop Button		

- **Writing of Ladder and Programming the PLC:**

Follow the procedure as shown in the Example # 1 for loading the Ladder Program & PT Screen.

In this program, LD input & OUT output and SFTL shift register commands are used and a simple latching sequence is used.